Case Study: AI speedup?

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Warmup

Work on the warmup activity (handout).

Warmup: designing a study

You have the opportunity to conduct a randomized experiment with a group of 16 experienced software developers, with the goal of assessing whether AI use decreases the time it takes the developers to complete tasks. How would you design this study? What would you measure?

Warmup: designing a study

A real 2025 study tried to answer this question.

Study participants: 16 experienced software developers, with moderate AI experience

Study design:

- authors identified 246 tasks in large, well-known projects on which the developers had an average of 5 years experience
- each task was randomly assigned to allow or disallow AI usage
- each task was assigned to one of the developers (each developer completed between 2 and 28 tasks)
- before being assigned the tasks, developers estimated how long each task would take with and without AI usage
- researchers then measured the time to complete each task

Activity part 2

Work on the next part of the activity (handout).

Proposed model

Researchers propose the following model:

$$\mathsf{T}_i = \beta_0 + \beta_1 \mathsf{AI}_i + \beta_2 \mathsf{EstTimeNoAI}_i + \varepsilon_i$$

Question: Are there any potential issues with this model? How would you propose modifying the model?

Adding a random intercept

$$\mathsf{T}_{ij} = \beta_0 + \beta_1 \mathsf{AI}_{ij} + \beta_2 \mathsf{EstTimeNoAI}_{ij} + u_i + \varepsilon_{ij}$$

Question: This model assumes that the effect of AI usage (β_1) is the *same* for all developers. How would I allow the effect of AI usage to *differ* between developers?

Adding a random slope

$$\mathsf{T}_{ij} = (\beta_0 + a_i) + (\beta_1 + b_i)\mathsf{AI}_{ij} + \beta_2 \mathsf{EstTimeNoAI}_{ij} + \varepsilon_{ij}$$

Adding a random slope

$$\mathsf{T}_{ij} = (\beta_0 + a_i) + (\beta_1 + b_i)\mathsf{AI}_{ij} + \beta_2 \mathsf{EstTimeNoAI}_{ij} + \varepsilon_{ij}$$

$$\varepsilon_{ij} \sim N(0, \sigma_\varepsilon^2) \qquad \begin{bmatrix} a_i \\ b_i \end{bmatrix} \sim N\left(\begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} \sigma_a^2 & \rho_{ab}\sigma_a\sigma_b \\ \rho_{ab}\sigma_a\sigma_b & \sigma_b^2 \end{bmatrix} \right)$$

Fitting the model in R

```
m1 <- lmer(initial_implementation_time ~
             ai_treatment + predicted_time_no_ai +
             (ai_treatment|dev_id), data = metr_ai)
```

```
Random effects:
 Groups   Name           Variance Std.Dev. Corr
 dev_id   (Intercept)    1205.2   34.72
          ai_treatment    352.9   18.79    -0.19
 Residual                5692.8   75.45
Number of obs: 246, groups:  dev_id, 16

Fixed effects:
                    Estimate Std. Error t value
(Intercept)        -18.57734   13.96529  -1.330
ai_treatment        17.36730   11.13830   1.559
predicted_time_no_ai  1.05437    0.07498  14.062
```

Activity part 3

Work on the next part of the class activity (handout).

Activity

Based on the diagnostics, which model would you choose?

If we chose model 1...

```
Fixed effects:
                       Estimate Std. Error t value
(Intercept)          -18.57734   13.96529  -1.330
ai_treatment          17.36730   11.13830   1.559
predicted_time_no_ai   1.05437    0.07498  14.062
```

How would you interpret a coefficient relevant to the research question?

If we chose model 2...

```
Fixed effects:
                      Estimate Std. Error t value
(Intercept)           -0.17707    0.28860  -0.614
ai_treatment           0.15577    0.09261   1.682
log(predicted_no_ai)   0.97549    0.06155  15.849
```

How would you interpret a coefficient relevant to the research question?

Inference

Explain how you would use parametric bootstrapping to further address the research question.